

On min-max polarization of sharp spherical codes that are not tight designs

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Abstract

We consider the dual of the classical polarization problem, i.e. we target minimization of the absolute maximum of (a large class of) discrete potentials over the unit sphere in $\mathbb{R}^n$. In order to solve the problem for sharp spherical codes that are not tight designs we introduce and investigate m -regular codes. This new property means that the $2m$ -power discrete potential of such codes is maximized at all their points. Our result completes the solution of the min-max polarization problem for all cases where sharp codes are known with one exception.

Key Words : Discrete potentials, Sharp spherical codes, Regular codes, Min-max polarization

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1 Introduction

Let $C = \{x_1, \dots, x_N\}$, $N \geq 2$, be a non-empty and finite subset of the unit sphere $\mathbb{S}^{n-1}$ in $\mathbb{R}^n$, $n \geq 2$, i.e. a spherical code.

For a given spherical code C and a function $h : [-1, 1] \rightarrow \mathbb{R}$, we define the h -potential of C by

$$U_h(x, C) := \sum_{i=1}^N h(x \cdot x_i), \quad (1)$$

where $x \cdot y$ is the standard dot product of vectors x and y in $\mathbb{R}^n$. We are interested in the solution to the following two problems.

Problem 1.1. (Maximum polarization problem) Given a spherical code $C \subset \mathbb{S}^{n-1}$, $n \geq 2$, find the quantity

$$Q_h(C, \mathbb{S}^{n-1}) := \sup_{x \in \mathbb{S}^{n-1}} \sum_{i=1}^N h(x \cdot x_i) \quad (2)$$

and points $x^* \in \mathbb{S}^{n-1}$ attaining the supremum on the right-hand side of (2), if they exist.

Problem 1.2. (Min-max polarization problem) Given $N \geq 1$, find the quantity

$$Q_h(N, \mathbb{S}^{n-1}) := \inf_{\substack{C \subset \mathbb{S}^{n-1} \\ \#C=N}} Q_h(C, \mathbb{S}^{n-1}) \quad (3)$$

and N -point codes $C \subset \mathbb{S}^{n-1}$ attaining the infimum on the right-hand side of (3), if they exist.

We will assume that the potential function h is absolutely monotone on $[-1, 1]$. This includes the important cases of Gaussian interactions $K(x, y) = e^{-a|x-y|^2}$, $a > 0$, where $h(t) = e^{-a(2-2t)}$, and Riesz interactions $K(x, y) = -|x-y|^{-s}$, $-2 < s < 0$ (after adding an appropriate constant), where $f(t) = (2-2t)^{-s/2}$. If $h(1)$ is allowed to be infinite, we can also consider the classical Riesz interactions with $s > 0$ and the logarithmic interactions $K_{\log}(x, y) = \frac{1}{|x-y|}$, where $f(t) = \frac{1}{2} \ln \frac{1}{2-2t}$ (again after adding an appropriate constant).

This setting allows using linear programming techniques to establish the universal optimality for Problems 1.1 and 1.2; i.e., optimality of codes with respect to all absolutely monotone potentials, using the relation to the energy minimization problem for Problem 1.2 (see (13) in the proof of Theorem 3.2). Problem 1.2 was solved in [Bor24] for the tight spherical designs. Here we extend this result by proving the universal optimality of sharp codes which are also regular (see Definition 2.1 below). This completes the solution of Problem 1.2 in all cases where sharp codes are known with one exception.

2 Preliminaries

A spherical code which is an m -distance set and a spherical $(2m - 1)$ -design for some positive integer m is called an m -sharp code [CK07]. If an m -sharp code is in fact $2m$ -design, it is called strongly sharp. If a spherical code is a $(2m - 1)$ -design forming m distinct dot products with some point on S^{n-1} , it is called m -stiff [Bor25]. The regular N -gon with N even, cube, cross-polytope, and 24-cell, are basic examples of stiff codes. Many interesting sharp and non-sharp codes are also stiff, see [Bor25, BDH⁺25].

For C being the set of vertices of a regular N -gon on S^1 or a regular simplex, or a regular cross-polytope, or a cube inscribed in S^{n-1} , the potential (1) with

$$h(t) = s(a - 2t)^{-s/2}, \quad s \neq 0, \quad a \geq 2, \quad (4)$$

(this includes the cases of Coulomb and Riesz potentials) was minimized and maximized over S^{n-1} (with the corresponding n) by Stolarsky [Sto75a, Sto75b] and by Nikolov and Rafailov [NR11, NR13]. Hardin, Kendall, and Saff [HKS13] found the minimizers of the potential of a regular N -gon inscribed in S^1 with respect to a decreasing and convex function of the geodesic distance (which includes potentials (4) with $s > -1$, $s \neq 0$, and $a = 2$).

For potentials (1) with $h : [-1, 1] \rightarrow (-\infty, \infty]$ which are finite and absolutely monotone on $[-1, 1)$ and continuous in the extended sense at $t = 1$, the points of absolute maxima coincide with the vertices of C when C is a regular simplex on S^{n-1} [Bor22], a strongly sharp code on S^{n-1} [Bor24], or an antipodal sharp code $C \subset S^{n-1}$ [Bor24]. Moreover, in all cases the points of absolute maximum appear to be the same for any strictly absolutely monotone potential function h on $[-1, 1)$. We call such points of absolute maximum *universal maxima* of the code C .

In this paper we extend this result by showing that the same pattern; i.e., points of the code being universal maxima, holds for a class of sharp codes which are not covered in [Bor22, Bor24]. To this end we introduce and investigate a new notion.

Definition 2.1. A spherical code $C = \{x_1, \dots, x_N\} \subset S^{n-1}$ is called *m -regular*, $n, N \geq 2, m \geq 1$, if the potential

$$U_{2m}(x, C) := \sum_{i=1}^N (x \cdot x_i)^{2m}$$

attains its absolute maximum over $x \in S^{n-1}$ at every point of C .

3 Optimality of m -sharp m -regular codes

The following result holds for universal maxima of the potential of sharp codes that satisfy Definition 2.1.

Theorem 3.1. *Let $n, N \geq 2$, $m \geq 1$, and a subset $C = \{x_1, \dots, x_N\} \subset \mathbb{S}^{n-1}$ be an m -sharp code. Assume, in addition, that C is m -regular. Then for every continuous potential function $h : [-1, 1] \rightarrow \mathbb{R}$ differentiable on $[-1, 1)$ such that $h^{(2m)} \geq 0$ and $h^{(2m+1)} \geq 0$ on $(-1, 1)$, the potential*

$$U_h(x, C) := \sum_{i=1}^N h(x \cdot x_i) \quad (5)$$

attains its absolute maximum over $\mathbb{S}^{n-1}$ at every point of C .

If, in addition, $h^{(2m)} > 0$ and $h^{(2m+1)} > 0$ on $(-1, 1)$, then no other point on $\mathbb{S}^{n-1}$ is a maximizer of the potential in (5).

Theorem 3.1 combined with a result by Cohn and Kumar [CK07] and the Delsarte-Yudin bound yield the following statement.

Theorem 3.2. *Let $n, N \geq 2$, $m \geq 1$, and $h : [-1, 1] \rightarrow \mathbb{R}$ be a continuous function differentiable on $[-1, 1)$ with $h^{(k)} \geq 0$ on $(-1, 1)$ for $1 \leq k \leq 2m + 1$. Suppose an N -point configuration $C^* \subset \mathbb{S}^{n-1}$ is m -sharp and m -regular. Then*

$$Q_h(C, \mathbb{S}^{n-1}) \geq Q_h(C^*, \mathbb{S}^{n-1}) \quad (6)$$

holds for every N -point multiset $C \subset \mathbb{S}^{n-1}$.

If, in addition, $h^{(k)} > 0$ on $(-1, 1)$ for $1 \leq k \leq 2m + 1$ and C is such that equality holds in (6), then C is an m -sharp set, the potential $U_h(\cdot, C)$ attains its absolute maximum over $\mathbb{S}^{n-1}$ at every point of C , and the same dot products occur in C and C^ .*

If C is strongly m -sharp, then it is m -regular, since the potential in Definition 2.1 is constant. Therefore, assertions of Theorem 3.1 and Theorem 3.2 hold. For strongly m -sharp codes, these results were earlier established by the author in [Bor24].

Results stated above can be applied and are, therefore, new to six m -sharp spherical codes that are not tight designs and are shown here to be m -regular for $m = 1, 2$, or 3 , see Table 1. They are solutions to the min-max polarization problem by Theorem 3.2 and the potential of each of them has its universal maxima at points of the code itself as asserted by Theorem 3.1.

4 Proof of Theorems 3.1 and 3.2

Before starting the proofs, we recall that a configuration C on $\mathbb{S}^{n-1}$ is m -stiff if it is a $(2m - 1)$ -design and there is a point on $\mathbb{S}^{n-1}$ that it forms at most m distinct dot products with points of C . Denote also by $\mathbb{P}_k$, $k \geq 0$, the set of all real polynomials of degree at most k .

n	N	Design strength	Dot products	Name/description
≥ 2	$2, \dots, n$	1	$-1/(N-1)$	lower-dimensional regular simplex
5	16	3	$-3/5, 1/5$	5-demicube/kissing of Schläfli code
21	112	3	$-1/3, 1/9$	a facet of McLaughlin code
21	162	3	$-2/7, 1/7$	kissing of McLaughlin code
22	100	3	$-4/11, 1/11$	Higman-Sims code
22	891	5	$-1/2, -1/8, 1/4$	kissing of 4600-pt. tight 7-design

Table 1: Sharp codes on $\mathbb{S}^{n-1}$ that are not tight designs for which we were able to show m -regularity (with $m = 1, 2$, or 3). The universal maxima of each of them are at points of the code via Theorem 3.1. These codes are universally optimal for the min-max polarization problem among all codes of the same cardinality via Theorem 3.2. Each of these codes is also m -stiff, for the corresponding $m = 1, 2$, or 3 , as observed in [BDH⁺25].

Proof of Theorem 3.1. Let the code C satisfies the condition of the theorem. Let $-1 \leq \alpha_1 < \dots < \alpha_m < 1$ be the distinct dot products between distinct points in C .

Let $p \in \mathbb{P}_{2m}$ be the polynomial that interpolates h and h' at the points $t = \alpha_i$, $i = 1, \dots, m$, and h at $t = 1$. Then, for every $t \in [-1, 1]$ there exists a point $c = c(t) \in (-1, 1)$ such that

$$h(t) - p(t) = \frac{h^{(2m+1)}(c)}{(2m+1)!} (t - \alpha_1)^2 \cdots (t - \alpha_m)^2 (t - 1). \quad (7)$$

Since $h^{(2m+1)} \geq 0$ on $(-1, 1)$, we have $h(t) - p(t) \leq 0$; that is $h(t) \leq p(t)$, $t \in [-1, 1]$. Let $q \in \mathbb{P}_{2m-1}$ be the polynomial such that $p(t) = a_{2m}t^{2m} + q(t)$. Let $g(t) := h(t) - p(t)$. By the Rolle's Theorem, g' has at least $2m$ zeros in $(-1, 1)$. Then $g^{(2m)}$ has at least one zero in $(-1, 1)$, which we will denote by ξ . Then

$$2m!a_{2m} = p^{(2m)}(\xi) = h^{(2m)}(\xi) \geq 0. \quad (8)$$

Consequently, $a_{2m} \geq 0$.

Since C is a $(2m-1)$ -design, the potential $U_q(x, C) = \sum_{i=1}^N q(x \cdot x_i)$ is constant over $x \in \mathbb{S}^{n-1}$. Let $k = 1, \dots, N$ be an arbitrary index. Then for every $x \in \mathbb{S}^{n-1}$,

since C is m -regular, we have consecutively

$$\begin{aligned}
U_h(x, C) &= \sum_{i=1}^N h(x \cdot x_i) \leq \sum_{i=1}^N p(x \cdot x_i) \\
&= a_{2m} \sum_{i=1}^N (x \cdot x_i)^{2m} + \sum_{i=1}^N q(x \cdot x_i) \\
&= a_{2m} \sum_{i=1}^N (x \cdot x_i)^{2m} + \sum_{i=1}^N q(x_k \cdot x_i) \\
&\leq a_{2m} \sum_{i=1}^N (x_k \cdot x_i)^{2m} + \sum_{i=1}^N q(x_k \cdot x_i) \\
&= \sum_{i=1}^N p(x_k \cdot x_i) = \sum_{i=1}^N h(x_k \cdot x_i) = U_h(x_k, C),
\end{aligned} \tag{9}$$

where the equality in the last line holds from the interpolation conditions. Therefore, the polarization potential $U_h(\cdot, C)$ attains its absolute maximum over $\mathbb{S}^{n-1}$ at the points x_k , $k = 1, \dots, N$, of C .

Assume now that $h^{(2m)} > 0$ and $h^{(2m+1)} > 0$ on $(-1, 1)$. Then $h(t) < p(t)$ for every $t \in [-1, 1] \setminus \{\alpha_1, \dots, \alpha_m, 1\}$. Assume for a contradiction that $U_h(\cdot, C)$ attains its maximum over $\mathbb{S}^{n-1}$ at a point in $\mathbb{S}^{n-1} \setminus C$. Denote such a point by z . Then, for example,

$$U_h(z, C) = U_h(x_1, C). \tag{10}$$

Then equality throughout the derivation (9) implies that $z \cdot x_i \in \{\alpha_1, \dots, \alpha_m\}$ for each $i = 1, \dots, N$. Since C is a $(2m-1)$ -design, the dot product $z \cdot x_i$ assumes every value $\alpha_1, \dots, \alpha_m$. Thus, the code is m -stiff (with respect to z). By Theorem 4.3 from [Bor25], the potential $U_h(\cdot, C)$ attains its absolute minimum over $\mathbb{S}^{n-1}$ at z . By the same theorem, since $h^{(2m)} > 0$, the potential $U_h(\cdot, C)$ does not attain its absolute minimum over $\mathbb{S}^{n-1}$ at any point on $\mathbb{S}^{n-1}$ forming more than m distinct dot products with points of C . Since x_1 forms $m+1$ distinct dot product with points of C (counting 1), it is not a point of minimum; that is, $U_h(z, C) < U_h(x_1, C)$ contradicting (10). Therefore, $U_h(\cdot, C)$ does not have maximizers in $\mathbb{S}^{n-1} \setminus C$. $\square$

The proof of Theorem 3.2 uses the following auxiliary statement, which follows from proofs in [CK07] (Lemma 4.1 below is not explicitly stated¹).

Lemma 4.1. *Let $n \geq 2$, $m \geq 1$, and $C^* \subset \mathbb{S}^{n-1}$ be an m -sharp code with dot products $-1 \leq t_1 < \dots < t_m < 1$ between its distinct points. If $h : [-1, 1] \rightarrow \mathbb{R}$ is differentiable with $h^{(\ell)} \geq 0$ on $(-1, 1)$ for $0 \leq \ell \leq 2m-1$, then the Hermite interpolating polynomial p for h at $t_1, \dots, t_m$ has non-negative ℓ -th Gegenbauer coefficients a_ℓ for $1 \leq \ell \leq m$.*

¹Paper [CK07] proves its results for absolutely monotone potentials and mentions that the assumptions on potential functions there can be relaxed to $2m$ - or $(2m-1)$ -monotonicity.

If, in addition, $h^{(\ell)} > 0$ on $(-1, 1)$, $1 \leq \ell \leq 2m - 1$, then $a_\ell > 0$, $1 \leq \ell \leq 2m - 1$.

Proof of Theorem 3.2. We recall the positive definiteness property of Gegenbauer polynomials (see, e.g., [BHS19, Chapter 5]). For any multiset $C = \{x_1, \dots, x_N\}$ on $\mathbb{S}^{n-1}$, we have

$$\sum_{i=1}^N \sum_{j=1}^N P_\ell^{(n)}(x_i \cdot x_j) \geq 0. \quad (11)$$

If $-1 \leq t_1 < \dots < t_m < 1$ are the dot products between distinct points in $C^* = \{x_1^*, \dots, x_N^*\}$ and p is the Hermite interpolating polynomial of degree $2m - 1$ for h at $t_1, \dots, t_m$, then for every $t \in [-1, 1]$, there is $c = c(t) \in (-1, 1)$ such that

$$h(t) - p(t) = \frac{h^{(2m)}(c)}{(2m)!} (t - t_1)^2 \cdots (t - t_m)^2 \geq 0. \quad (12)$$

Therefore, $h(t) \geq p(t)$, $t \in [-1, 1]$.

Let $p(t) = \sum_{\ell=0}^{2m-1} a_\ell P_\ell^{(n)}(t)$ be the Gegenbauer expansion of p . Then we have $a_\ell \geq 0$, $1 \leq \ell \leq 2m - 1$, by Lemma 4.1. Using the positive definiteness property (11), we write the Delsarte-Yudin bound:

$$\begin{aligned} Q_h(C, \mathbb{S}^{n-1}) &\geq \max_{i=1, N} \sum_{j=1}^N h(x_i \cdot x_j) \geq \frac{1}{N} \sum_{i=1}^N \sum_{j=1}^N h(x_i \cdot x_j) \\ &= h(1) + \frac{1}{N} \sum_{1 \leq i \neq j \leq N} h(x_i \cdot x_j) \\ &\geq h(1) - p(1) + \frac{1}{N} \sum_{1 \leq i, j \leq N} p(x_i \cdot x_j) \\ &= h(1) - p(1) + \frac{1}{N} \sum_{\ell=0}^{2m-1} a_\ell \sum_{1 \leq i, j \leq N} P_\ell^{(n)}(x_i \cdot x_j) \\ &\geq h(1) - p(1) + a_0 N. \end{aligned} \quad (13)$$

If we let $C = C^*$, then by Theorem 3.1 we have equality throughout the first line in (13). Since p interpolates f at every value of the dot products $x_i^* \cdot x_j^*$, $i \neq j$, we have equality in the second line of (13). Finally, since C^* is a $(2m - 1)$ -design, we have equality in (11) for each $\ell = 1, \dots, 2m - 1$. This gives equality in the third line of (13) thus proving (6).

Assume that $h^{(k)} > 0$ on $(-1, 1)$ for every $k = 1, \dots, 2m + 1$ and that C is such a multiset on $\mathbb{S}^{n-1}$ that equality holds in (6). Then equality holds throughout (13). Assume for a contradiction that C possesses repeating points; i.e. some distinct points x_i, x_j in C form dot product 1. Then (12) and the fact that $h^{(2m)} > 0$ on $(-1, 1)$ imply that $p(t) < f(t)$ on $[-1, 1] \setminus \{t_1, \dots, t_m\}$. In particular, $p(1) <$

$h(1)$. Then we have a strict inequality in the second line of (13) contradicting the assumption that equality holds in (6). Thus, $x_i \neq x_j$ whenever $i \neq j$ in C .

By Lemma 4.1, we have $a_\ell > 0$, $\ell = 1, \dots, 2m-1$. Then equality throughout (13) implies equality in (11) for all $\ell = 1, \dots, 2m-1$; i.e., C is a $(2m-1)$ -design. Since $p(t) < h(t)$ for $t \in [-1, 1] \setminus \{t_1, \dots, t_m\}$, the equality between the second and third lines of (13) implies that $x_i \cdot x_j \in \{t_1, \dots, t_m\}$, $i \neq j$; that is, C is m -sharp. Since C is a $(2m-1)$ -design, the dot products $x_i \cdot x_j$ assume every value $t_1, \dots, t_m$. Finally, the equalities in the first line of (13) imply that the potential $U_h(\cdot, C)$ attains its absolute maximum over S^{n-1} at every point of C . $\square$

5 Tools for proving the regularity of spherical codes

We need the concepts of weighted spherical designs [GS81] and derived codes [DGS77].

A spherical code $C \subset S^{n-1}$ is called a weighted spherical k -design, $k \geq 0$, if for each $y \in C$, there exists a positive number w_y such that for every polynomial $p \in \mathbb{P}_k$,

$$\sum_{y \in C} w_y p(x \cdot y) = a_0^n(p) \sum_{y \in C} w_y, \quad x \in S^{n-1},$$

where $a_0^n(p) = \int_{-1}^1 p(t) w_n(t) dt$ is the zeroth Gegenbauer coefficient of $p(t)$ corresponding to S^{n-1} and $w_n(t) := \gamma_n (1-t^2)^{\frac{n-3}{2}}$ is the Gegenbauer weight with the positive constant γ_n selected to make $w_n(t)$ a probability density on $[-1, 1]$.

For a spherical code $C \subset S^{n-1}$, $n \geq 3$, and a number $\alpha \in (-1, 1)$, denote by

$$C(\alpha) := \left\{ x \in S^{n-2} : \left(\sqrt{1-\alpha^2} x, \alpha \right) \in C \right\}$$

and by $\overline{C}(\alpha) \subset S^{n-1}$ the set of points in C forming dot product α with vector $\mathbf{x} := (0, \dots, 0, 1) \in \mathbb{R}^n$. The code $C(\alpha) \subset S^{n-2}$ is called derived of C [DGS77]. Denote by $I(x, C)$ the set of all distinct values of the dot product formed by a given point $x \in S^{n-1}$ with points from C . Let also $\epsilon = 1$ if $\mathbf{x} \in C$ and $\epsilon = 0$ otherwise and let $\delta = 1$ if $-\mathbf{x} \in C$ and $\delta = 0$ otherwise.

We present without proofs the results which are needed for our m -regularity proofs for the codes in Table 1.

Theorem 5.1. *Let $C \subset S^{n-1}$, $n \geq 3$, be a spherical $(k+1)$ -design, $k \geq 0$, and let the set $I(\mathbf{x}, C) \cap (-1, 1)$ consist of dot products $-1 < \alpha_1 < \dots < \alpha_\ell < 1$. Assume that $C(\alpha_i)$ is a k -design on S^{n-2} , $i = 1, \dots, \ell$. Then the set $\mathbf{C} := C(\alpha_1) \cup \dots \cup C(\alpha_\ell) \subset S^{n-2}$, where each point $y \in \mathbf{C}$ has weight $w_y = \sum_{i \in J_y} (1 - \alpha_i^2)^{\frac{k+1}{2}}$, and $J_y = \{i : C(\alpha_i) \ni y\}$, is a weighted $(k+1)$ -design on S^{n-2} .*

We remark that the set J_y may contain more than one index, since subconstituents $C(\alpha_i)$ may intersect or coincide for different i .

We next give a sufficient condition for a pair of stiff configurations to consist of regular codes. We need the following special cases (constructions) of weighted designs.

Definition 5.2. We say that a pair (C, C_1) of configurations on $\mathbb{S}^{n-1}$, $n \geq 2$, forms a *weighted $2m$ -design*, $m \geq 1$, if there exist numbers $w, w_1 > 0$ such that

$$w \sum_{y \in C} P_k^{(n)}(x \cdot y) + w_1 \sum_{y \in C_1} P_k^{(n)}(x \cdot y) = 0, \quad x \in \mathbb{S}^{n-1}, \quad k = 1, \dots, 2m. \quad (14)$$

Lemma 5.3. Let $n \geq 2$, $m \geq 1$, and D and E be two $(2m - 1)$ -designs on $\mathbb{S}^{n-1}$ such that $\beta^2 - \alpha\gamma = 0$, where

$$\alpha := \sum_{x \in D} \sum_{y \in D} P_{2m}^{(n)}(x \cdot y) > 0, \quad \gamma := \sum_{x \in E} \sum_{y \in E} P_{2m}^{(n)}(x \cdot y),$$

and $\beta := \sum_{x \in D} \sum_{y \in E} P_{2m}^{(n)}(x \cdot y) < 0$. Then the pair (D, E) forms a *weighted $2m$ -design* on $\mathbb{S}^{n-1}$.

Given a spherical code $C \subset \mathbb{S}^{n-1}$, we denote by $\mathcal{D}_m(C)$, $m \geq 1$, the set of all points on $\mathbb{S}^{n-1}$ forming at most m distinct dot products with points of C . The following statement holds.

Lemma 5.4. Let $n \geq 2$ and $m \geq 1$. Suppose that (C, C_1) is a pair of $(2m - 1)$ -designs on $\mathbb{S}^{n-1}$ that forms a *weighted $2m$ -design* in the sense of Definition 5.2 such that $C_1 \subset \mathcal{D}_m(C)$ and $C \subset \mathcal{D}_m(C_1)$. Then C and C_1 are both *m -regular*.

6 Regularity of codes from Table 1

The proof of 1-regularity of the low dimensional regular simplexes from Table 1 is, more or less, direct. For the other four sharp codes, we need a more involved argument using the results from the previous section.

The sharp codes from Table 1 are derived codes of larger sharp codes. We will assume that these higher dimensional sharp codes, say C_{27} , C_{275} , C_{552} , and C_{4600} (all these are tight spherical designs), contain the point $\mathbf{x} = (0, \dots, 0, 1)$ in the space of corresponding dimension. Then the sharp codes from Table 1 arise as their derived codes. We shall use the notation of Theorem 5.1.

We present the proof of the 2-regularity of the sharp codes with 112 and 162 points on $\mathbb{S}^{20}$. The 16-point, 100-point, and 891-point sharp codes can be dealt similarly.

There exists a strongly 2-sharp code C_{275} on S^{21} with $N = 275$ points (McLaughlin code; a tight spherical 4-design). Within this code, the dot products are $\alpha_1 = 1/6$ (162 points of C_{275}) and $\alpha_2 = -1/4$ (112 points of C_{112}), the subconstituent (derived code) $C(\alpha_1)$ is the 162-point 2-sharp code on S^{20} and the subconstituent

$C(\alpha_2)$ is the 112-point 2-sharp code on S^{20} (see, e.g., [CK07, BDH⁺25]). The code $C(\alpha_1)$ (properly scaled) is the set of nearest neighbors for a vertex of the McLaughlin configuration. By Theorem 5.1, the codes $C(\alpha_1)$ and $C(\alpha_2)$ form a weighted 4-design on S^{20} . Since only two dot products can occur within $\mathcal{C}_{275}$, we have $C(\alpha_2) \subset \mathcal{D}_2(C(\alpha_1))$ and $C(\alpha_1) \subset \mathcal{D}_2(C(\alpha_2))$. Since both codes are 2-sharp, they are 3-designs. By Lemma 5.4, both codes are 2-regular. Then the assertions of Theorems 3.1 and 3.2 hold with $m = 2$ and $n = 21$ for both $C(\alpha_1)$ and $C(\alpha_2)$.

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